

Mine subsidence monitoring & early warning

Problem Statement Title:-

Theme: Smart Automation

Team Name: MineNova6

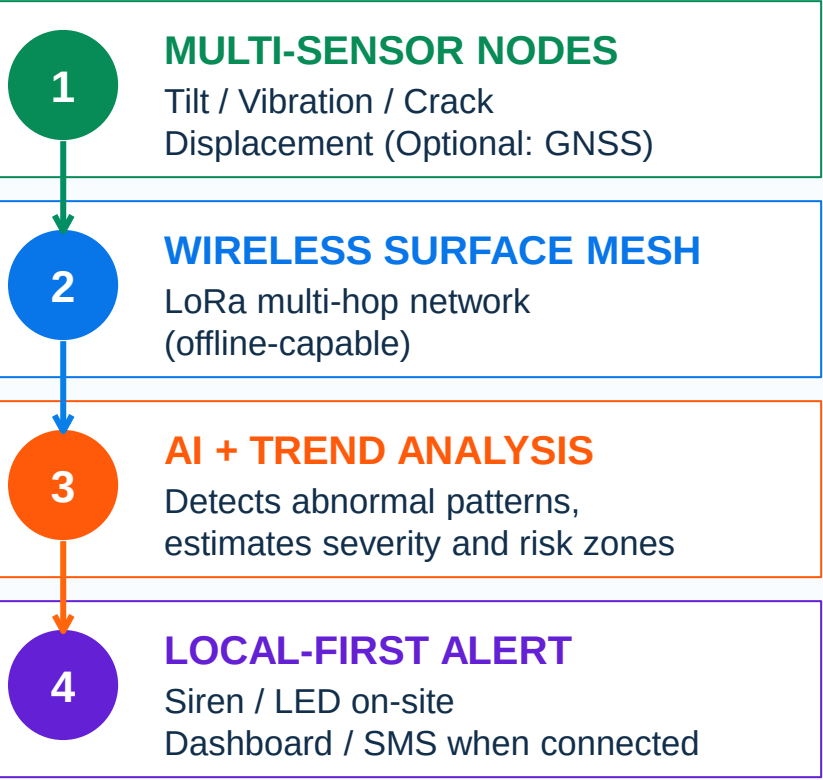
Team ID: 150683



Proposed Solution (Describe your Idea / Solution / Prototype)

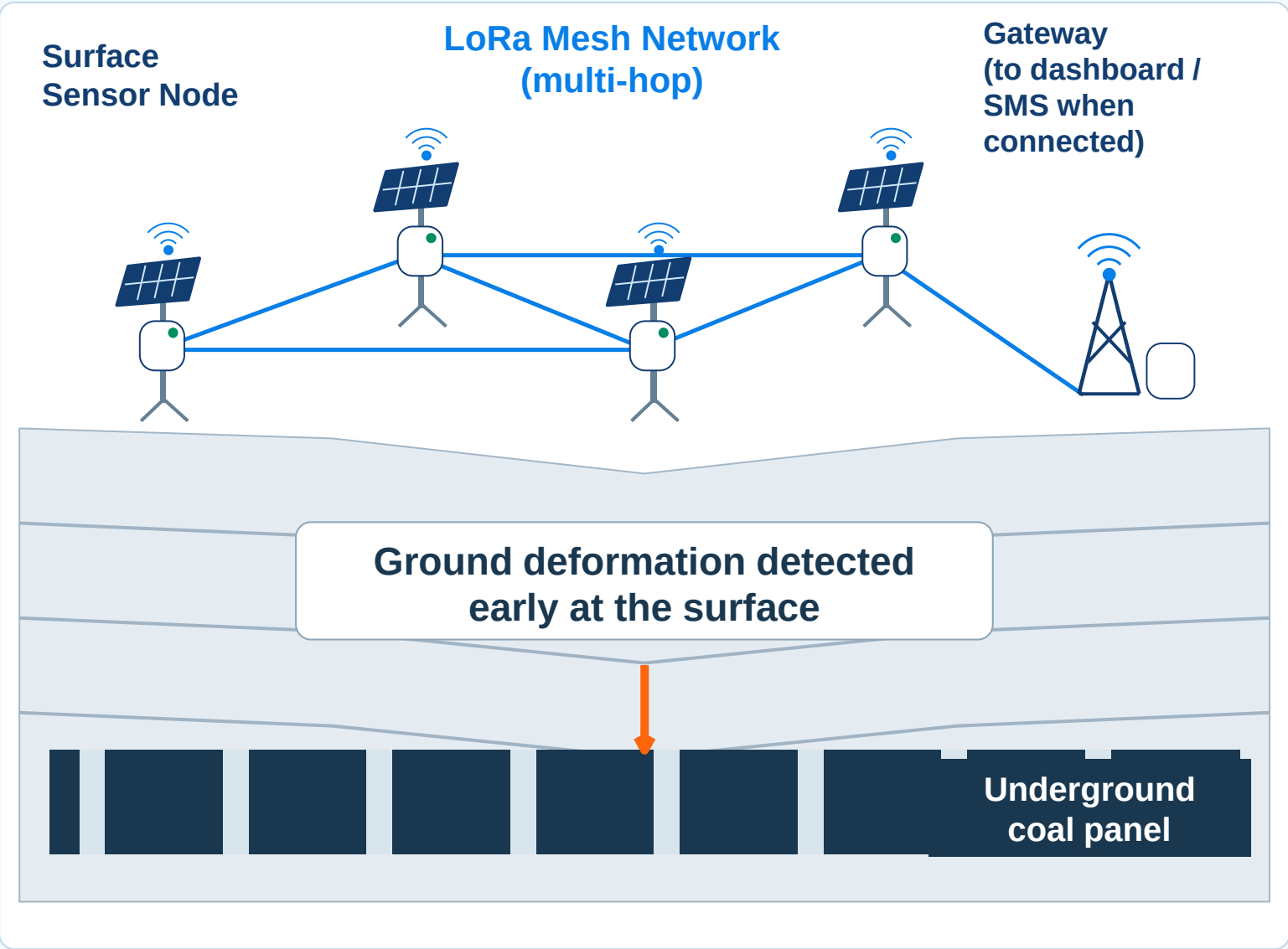
What is SubsiGuard?

SubsiGuard is a low-cost, surface-deployed early warning system that continuously monitors ground deformation above underground coal panels using a wireless sensor mesh, AI-based analysis and local-first alerts to enable timely risk mitigation.



Surface Sensors. Deeper Safety.

A wireless mesh network deployed on the surface above underground coal panels



The Existing Gap

Current Approach	Key Limitation	Our Advantage
Manual levelling	Periodic measurements	Continuous real-time monitoring
GNSS / Total station	High field cost	Low-cost distributed nodes
Satellite InSAR	Revisit delay	Local real-time indication
Vendor IoT systems	Connectivity dependency	Offline local mesh + on-site siren



THE MISSING LAYER

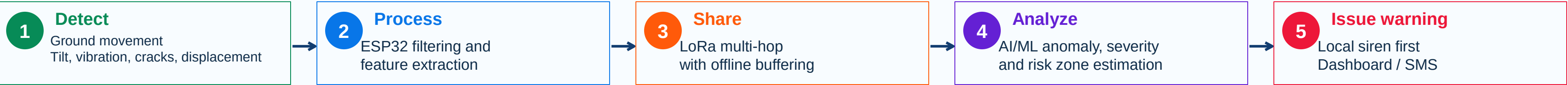
Affordable + Continuous + Local +
Offline-Capable Early Warning



SubsiGuard Fills This Gap

An indigenous, low-cost, scalable wireless surface mesh network designed for Indian coal mines.

How It Works? (End-to-End Flow)



1 Sense at Surface

Tilt | Vibration
Crack | Displacement
(Optional: GNSS)

2 Process at Node

ESP32
Filtering, feature extraction
Local threshold (fail-safe)

3 Transmit through Mesh

LoRa
Multi-hop mesh network
Offline buffering

4 Analyze & Predict

AI/ML (Random Forest)
Inverse velocity analysis
Risk zone mapping

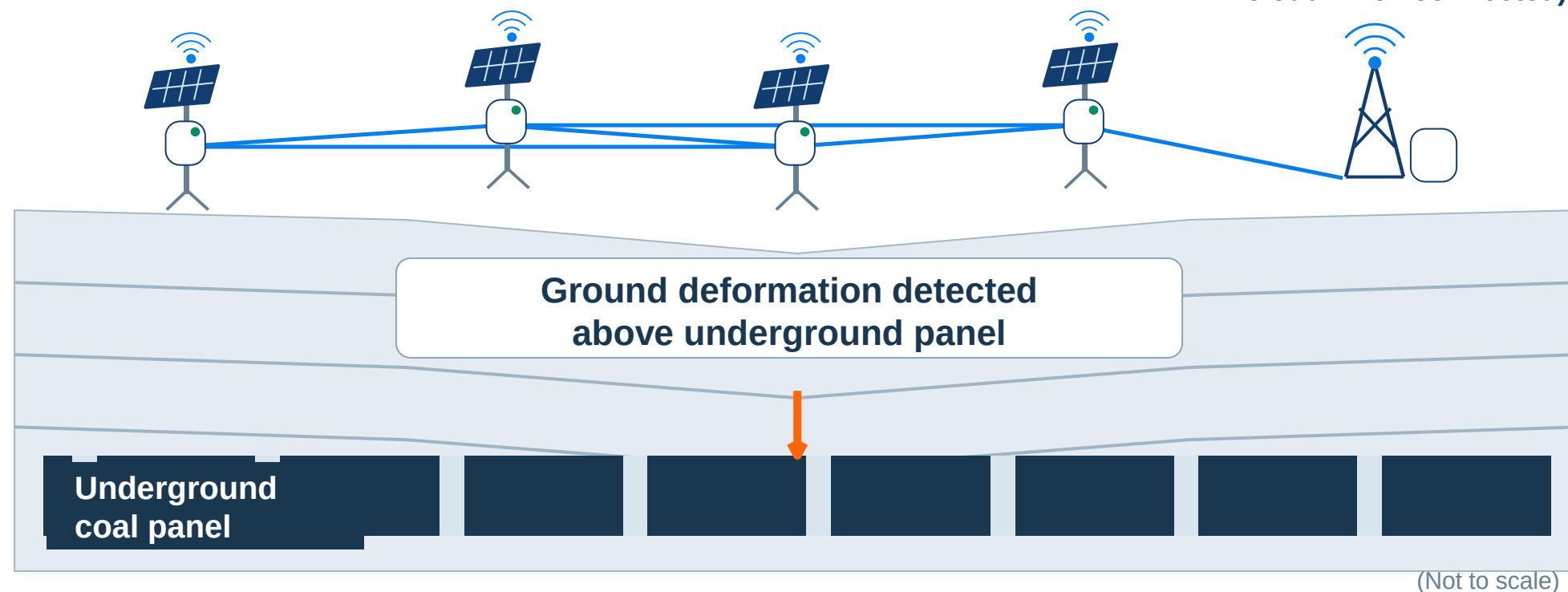
5 Generate Alerts

Local siren (works offline)
Dashboard / GIS
SMS / Email

Outputs: GIS Risk Map
Local Siren
SMS / Email

Surface Deployment

(Above underground coal panels)

Surface Deployment
(Above underground coal panels)**LoRa Mesh Network**
(Multi-hop, Self-healing)**Gateway**
(to dashboard /
cloud when connected)**Data Flow & System Architecture****1 Sensor Node (x N)**

MEMS Tilt Sensor
Vibration Sensor
Crack / Strain Sensor
Ultrasonic / Displacement
Optional: GNSS

2 LoRa Mesh Network

865–868 MHz, WPC compliant

3 ESP32 Gateway

Local server + Internet when available

4 AI Analysis

Random Forest
Anomaly detection
Severity estimation

5 Trend Analysis

Inverse Velocity
Progression
estimation
Early failure indication

System Outputs (Prototype / Web)**Live Sensor Dashboard**

Tilt 0.12° Vibration 0.03 g
Displacement 2.4 mm Crack Width 0.1 mm
Status: Normal1
*Illustrative Prototype Data

GIS Risk Zone Map**High Risk Detected**

Node N3 · Vibration + Displacement ↑
Risk Level: CRITICAL
Sijua Coalfield, Dhanbad
14:32 | 30 Sep 2026

1**Designed for Indian Conditions**

865–868 MHz (WPC band)
Low power & solar assisted
Rugged and field deployable

2**Offline-First Operation**

Works without internet
Local data buffering
Periodic sync when connected

3**Scalable Architecture**

Add more nodes easily
Multi-coalfield deployment
Self-healing mesh network

4**Low-Cost Prototype**

~ ₹2,100 per node (approx.)
Open-source stack
Student prototype friendly

5**Supports Smarter Decisions**

Real-time risk visibility
Helps evacuation & inspections
Protects workers and communities

FEASIBILITY & VIABILITY

Practical. Affordable. Deployable. Made for India.

Hardware Components (Per Sensor Node)

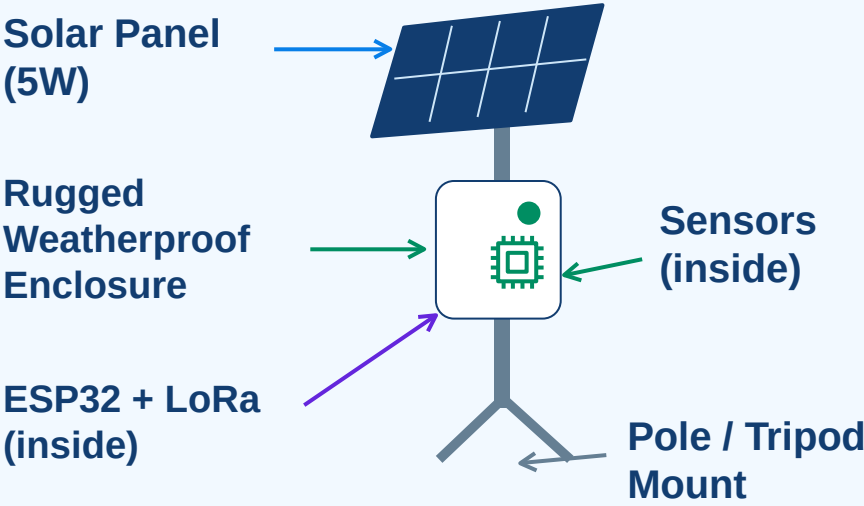
Built using readily available and affordable components

Component	Model / Specification	Approx. Cost (₹)
ESP32 Dev Module	ESP32-WROOM-32	450
Tilt Sensor	MPU6050 (IMU)	180
Vibration Sensor	SW-420	70
Displacement Sensor	String Potentiometer / Ultrasonic (HC-SR04)	250
Crack Sensor	Strain Gauge Module	200
LoRa Module	SX1276 (865–868 MHz)	350
GPS (Optional)	Neo-6M	200
Power Management	TP4056 + Lithium Battery	250
Solar Panel (5W)	+ Enclosure	200
Total (Approx.)		~ ₹2,100 per node

₹ **Low-cost, open-source hardware stack makes it student-prototype friendly and scalable.**

Prototype Sensor Node

Compact. Rugged. Field Deployable.



Low Power Design
Easy Deployment

Rugged & Weatherproof
Uses Readily Available Components

From Prototype to Large-Scale Deployment

A realistic and scalable path for real-world implementation

1	Student Prototype	Build & test individual node Validate sensors Basic mesh communication
2	Controlled Validation	Test in campus/open field Data collection Refine thresholds
3	Outdoor Field Test	Multi-node mesh Real environment testing Evaluate range and reliability
4	Pilot in Coal Mines	Deploy in selected mine site Monitor real data Work with mine operators
5	Scale Across Coalfields	Multi-site deployment Central dashboard Periodic sync Policy integration

Challenges in Indian Mining Conditions

Challenge	Our Mitigation
Dust, heat, humidity	Rugged IP-rated enclosure, industrial-grade components
Monsoon and water ingress	Sealed enclosures, proper mounting, water-resistant design
Unreliable connectivity	Offline mesh network + local buffering + periodic sync
Power availability	Solar-assisted, low-power design, sleep modes
False alarms / noise	Multi-sensor data fusion, AI/ML + threshold fail-safe
Harsh terrain & accessibility	Portable, easy-to-install node design
High deployment cost	Low-cost hardware, scalable mesh, mass deployment using local vendors

Power & Communication

- Solar-assisted operation**
- 5W solar panel
 - Li-ion battery
 - Low power mode
 - Long deployment life
- LoRa (865–868 MHz)**
- Long range
 - Multi-hop mesh
 - WPC compliant
 - Works in remote areas

Deployment Cost (Illustrative Estimate)

Item	Approx. Cost (₹)
Sensor Node (per unit)	~ 2,100
Gateway (per site)	~ 8,000 – 12,000
Installation (per node)	~ 500 – 1,000
Annual Maintenance (per node)	~ 300 – 500

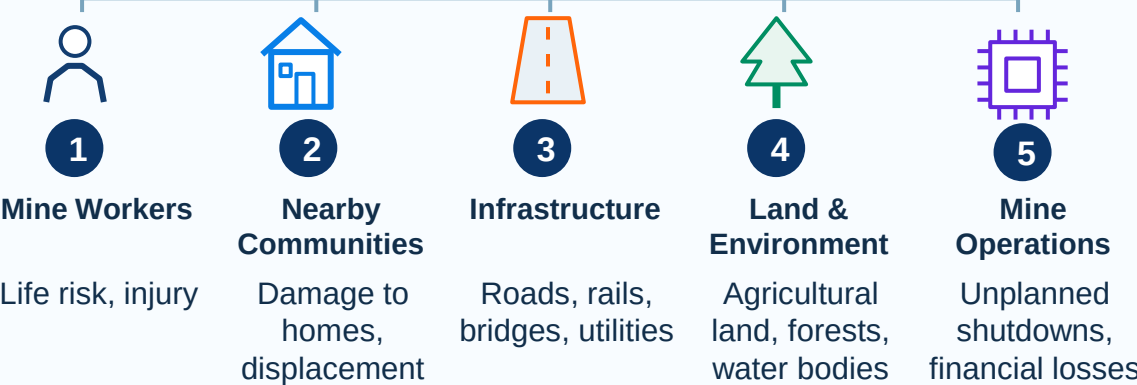
Why It's Feasible: Readily available components Low cost (~ ₹2,100/node) Low power & solar-assisted Easy to deploy and maintain Scalable for multiple coalfields Suited for Indian geo-mining conditions

IMPACT & BENEFITS

Early Detection. Safer Lives. Stronger Communities. Sustainable Mining.

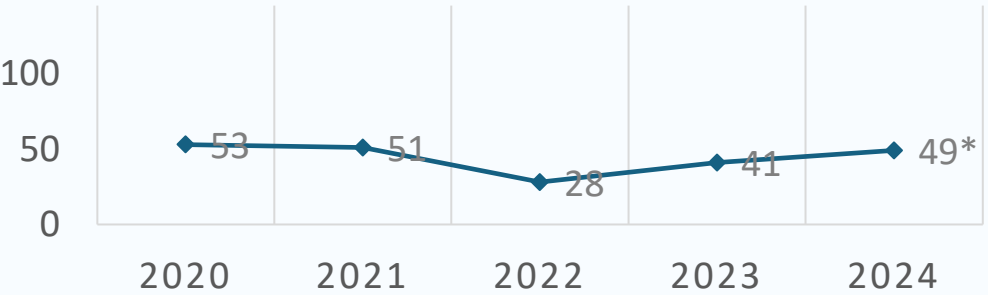
The Real-World Impact

Subsidence can affect far more than just the mine.



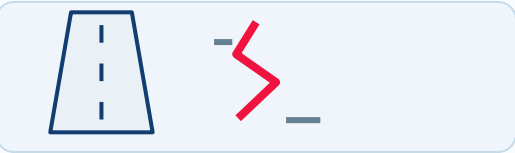
COAL MINE FATALITIES IN INDIA

All causes, DGMS report



Recent Subsidence-Related Incidents

Ground subsidence has caused damage to infrastructure, land and nearby habitations in multiple Indian coalfields.



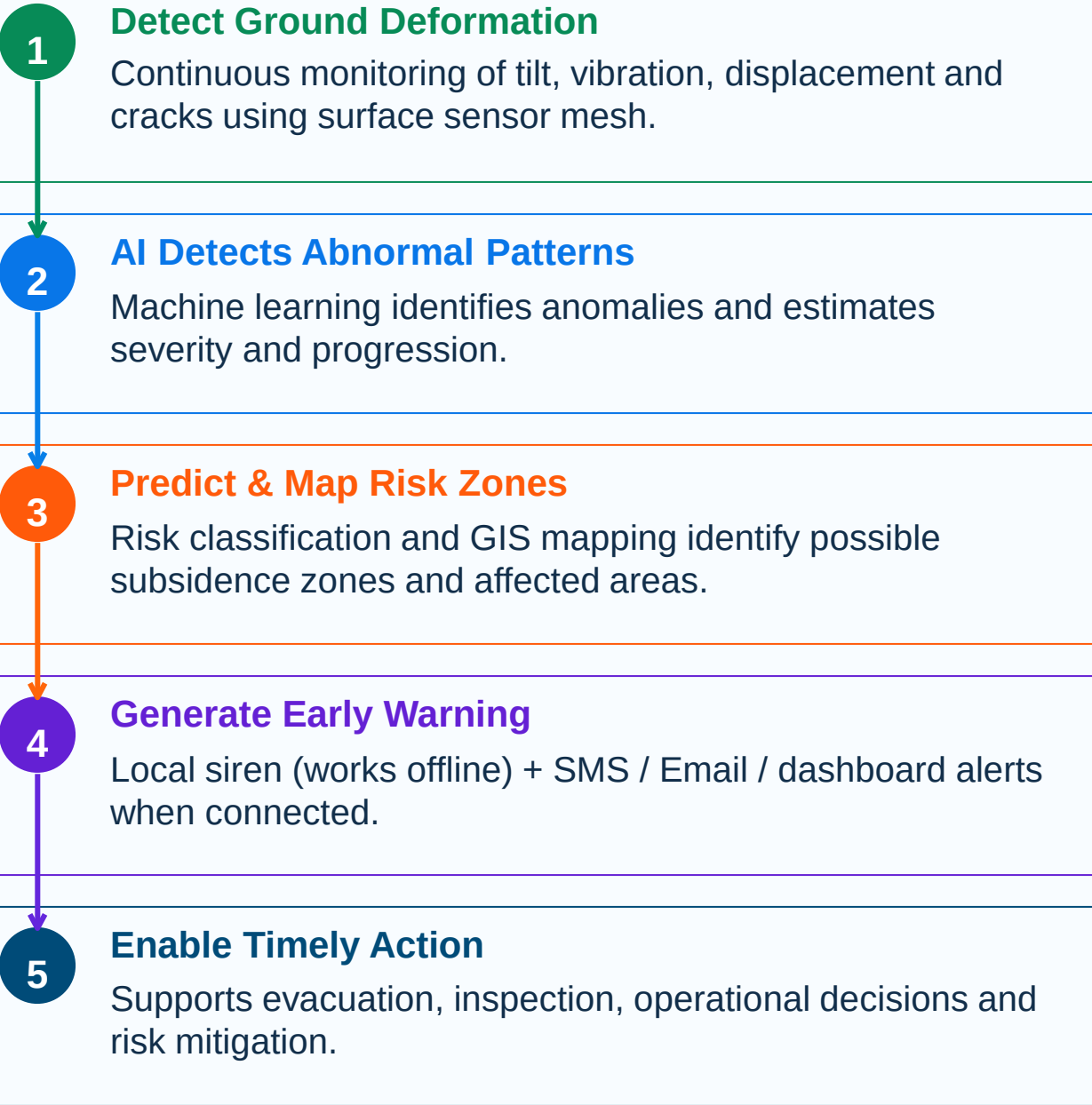
Jharia, Jharkhand (2022)
Roads and buildings damaged due to ground subsidence.



Raniganj, West Bengal (2019)
Houses developed cracks and tilted due to subsidence.

How SubsGuard Creates Impact

From ground movement detection to timely action.



Key Benefits

A safer, more resilient and sustainable mining ecosystem.



Aligned with National Goals

Safer Mines **Sustainable Development** **Disaster Risk Reduction** **Atmanirbhar Bharat**

Our Mission
To prevent tomorrow's disaster by detecting today's warning signs.

No fixed lives-saved percentage is claimed before field validation. Our goal is real impact through early detection and timely action.

*Coal-mine fatalities (all causes), India | Source: DGMS Annual Report 2024 | 2024 provisional

Literature-backed approach

Indian safety & mining policies

Proven technologies

Official data & regulations

1. Directorate General of Mines Safety (DGMS) — Annual Report 2024

Official coal-mine accident and fatality statistics. [DGMS](#)

2. Ministry of Coal — Annual Report 2023–24

Official information on coal mining, sustainability and mine safety. The Ministry's report specifically contains sections for both **Sustainability in Coal Mines** and **Safety in Coal Mines**.
[Ministry of Coal](#)

3. Coal Mines Regulations, 2017 — DGMS

Official Indian statutory framework governing coal-mine safety. DGMS currently lists the regulations in its mine-safety legislation repository. [DGMS](#)

4. Karanam et al. (2021) — “Multi-sensor remote sensing analysis of coal fire induced land subsidence in Jharia Coalfields, Jharkhand, India”

International Journal of Applied Earth Observation and Geoinformation, Vol. 102, Article 102439. This directly supports the fact that mine-related subsidence in Jharia can be measured and spatially mapped. [ScienceDirect](#)

5. Riyas et al. (2021) — “Detecting and Analyzing the Evolution of Subsidence Due to Coal Fires in Jharia Coalfield, India Using Sentinel-1 SAR Data”

Remote Sensing, 13(8), 1521. This is another directly relevant Indian coalfield subsidence-monitoring study.

6. Saito (1965) & Fukuzono (1985): Inverse-Velocity Model ($\frac{1}{v} \rightarrow 0$) for Tertiary Creep Failure Forecasting.

7. DGMS & CSIR-CIMFR Guidelines: Permissible Indian coalfield limits for surface tilt ($<0.57^\circ$) and tensile strain ($<5 \text{ mm/m}$).

8. Knothe (1957) & S.S. Peng (SME): Empirical Strata Movement & Surface Subsidence Engineering.

PROJECT RESOURCES

subsi-guard-ai.vercel.app

github.com/sharequesyed/SubsiGuardAI

Key Takeaways from Research

- Multiple low-cost sensor and remote monitoring methods are proven and feasible.
- AI/ML can effectively detect abnormal patterns for early warning.
- Wireless mesh networks and solar-powered IoT systems are field-deployable.
- Our solution aligns with national goals for safe, sustainable and technologically advanced mining.

Note on Data & Claims

- All statistics are sourced from official reports, peer-reviewed papers and credible media sources.
- We do not claim fixed lives-saved percentages before field validation.
- This project is a student prototype, designed to prove feasibility and demonstrate real-world potential.